

Arystan Shokan

Applied Scientist / ML Engineer | Optimization · Graph ML · Spatial Data

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Summary

Applied mathematician and engineer with an M.A. in Mathematics (Boston University) and 4+ years of industry experience turning mathematical models into production systems in combinatorial optimization, spectral and graph methods, spatial data, and probabilistic state estimation.

Technical Skills

ML & Data:	PyTorch, scikit-learn, Keras, KotlinDL, NumPy, Pandas, Matplotlib, SQL
Optimization:	Linear Programming, Dynamic Programming, Greedy & Heuristic Methods, Genetic Algorithms
Graph & Spatial:	Spectral Graph Methods, Graph Augmentation, R-trees, Computational Geometry
Probabilistic:	Kalman Filtering, Prediction–Correction Estimation, Statistical Modeling, Multilateration
Languages & Tools:	Python, Kotlin, C/C++, Julia, Java, Git, Docker
Mathematics:	Linear Algebra, Probability & Statistics, Differential Geometry, Algebraic Topology, Graph Theory

Experience

Engineer-Mathematician

04/2026 – Present

NaviX

Astana, Kazakhstan

- Design graph optimization algorithms (linear/dynamic programming, greedy methods) for parking selection
- Model the problem domain with statistical manifolds to represent high-dimensional data relationships
- Encode spatial data with R-trees and convert it into graph representations for efficient querying and analysis
- Apply spectral methods for subgraph selection, extracting the relevant network structures for optimization
- Implemented a graph augmentation algorithm that improves connectivity and downstream optimization results

Research and Development Engineer

10/2022 – 12/2023

Innoforce Systems

Almaty, Kazakhstan

- Stabilized GPS positioning by combining Android sensor and location-provider signals, reaching **up to 5 m accuracy**
- Implemented Kalman Filter and prediction–correction estimation, plus WiFi-RTT multilateration
- Deployed the algorithms on resource-constrained Android 7 payment terminals in city buses

Android Developer

02/2024 – 12/2024

Bereke Bank

Almaty, Kazakhstan

- Core platform team for the B-Business banking app; functional programming practices **cut runtime errors by 40%**, and architecture work **reduced legacy code by 70%**

Kotlin Multiplatform Developer

11/2020 – 09/2022

Kazdream Technologies

Astana, Kazakhstan

- Designed a graph-theoretic clustering algorithm for map data (**10× fewer rendered elements**) and **cut render time from 3 s to 3 ms** with reactive programming and caching

Projects

Vehicle Routing Problem Solver & Visualizer | *Kotlin, Linear Programming, Genetic Algorithms*

- Solves VRP instances by combining exact LP formulations with genetic-algorithm search, with live route visualization
- Wrote an accompanying paper whose literature review covers learning-based approaches, including graph neural networks

Movie Recommendation App | *Kotlin, Jetpack Compose, MVI, Coroutines, Flow*

- Recommends movies by cosine similarity between vector representations, served in a native Android client

Education & Research

M.A. in Mathematics (GPA 3.8/4.0)

01/2025 – 05/2026

Boston University

Boston, MA, USA

- Expository paper on *combinatorial Hodge theory* (smooth, discrete, and matroid): Hodge Laplacians on graphs and simplicial complexes, the mathematical basis of spectral and topological graph learning
- Expository paper on geodesics in Riemannian geometry, with a Python/OpenGL geodesic visualizer

B.Sc. in Software Engineering with Honors (GPA 3.8/4.0), Valedictorian

09/2020 – 06/2023

Astana IT University · Math exchange, Kyungpook National University (4.2/4.3)

Astana, Kazakhstan